Comprehensive Analytical Models for ^{87}Rb Trapping, Adiabatic Compression, Moving Lattice Transport, and EIT Cooling

Experimental Physics Laboratory Notes

June 6, 2026

Contents

1 Experimental Setup: Single Optical Dipole Trap (ODT)

1.1 Initial Parameters at the MOT Position

The initial state consists of a cold, spherical Magneto-Optical Trap (MOT) of ^{87}Rb containing $N \sim 10^8$ atoms with a cloud radius of $R \sim 1$ mm at an equilibrium temperature of $T = 8\,\mu$K.

The atoms are loaded into a free-fall Optical Dipole Trap (ODT) driven by a far red-detuned laser with total power $P = 1$ W and wavelength $\lambda = 1064$ nm. As the atoms fall over a distance of $d = 7$ mm, the laser beam waist focuses down from $W_{\text{MOT}} = 126\,\mu$m at the center of the MOT to $W_0 = 19\,\mu$m at the top of the Photonic Crystal Fiber (HCPCF).

For a laser beam propagating along the z-axis, the maximum transverse intensity at the center ($r = 0$) is defined by:

$$I_0 = \frac{2P}{\pi W^2} \tag{1}$$

Substituting the experimental parameters at the MOT trapping position ($P = 1$ W and $W_{\text{MOT}} = 126\,\mu$m), we obtain:

$$I_{\text{MOT}} = \frac{2 \cdot 1}{\pi (126 \cdot 10^{-6})^2} \simeq 4.01 \cdot 10^7 \text{ W/m}^2 \tag{2}$$

1.2 Derivation of the Dipole Potential

For a far-detuned laser, the optical dipole potential U_0 experienced by an atom is determined by its ground-state polarizability. For an idealized two-level atomic system, the potential scales as:

$$U = \frac{3\pi c^2}{2\omega_0^3} \left(\frac{\Gamma}{\Delta} \right) I_0 \tag{3}$$

Accounting for the fine structure of the D_1 and D_2 transitions of ^{87}Rb, the multi-level dipole potential formula yields:

$$U(z) = \pi c^2 \left(\frac{\Gamma_1}{\omega_1^3 \Delta_1} + \frac{2\Gamma_2}{\omega_2^3 \Delta_2} \right) I_0(z) \tag{4}$$

Substituting $I_0(z) = \frac{2P}{\pi W^2(z)}$, the full spatial dependency of the trap depth is given by:

$$U(z) = \pi c^2 \left(\frac{\Gamma_1}{\omega_1^3 \Delta_1} + \frac{2\Gamma_2}{\omega_2^3 \Delta_2} \right) \frac{2P}{\pi W^2(z)} \tag{5}$$

Evaluating this expression at the initial MOT coordinate where $W = 126\,\mu$m, the potential depth in temperature units is calculated as:

$$\frac{U_{\text{MOT}}}{k_B} \sim 6\,\mu\text{K} \tag{6}$$

2 Beam Propagation and Geometry at the Fiber Tip

2.1 Rayleigh Range and Waist Verification

We verify the consistency of the beam geometry assuming a focal waist $W_0 = W_{\text{tip}} = 19\,\mu\text{m}$ at the fiber tip and $\lambda = 1064$ nm. The Rayleigh range z_R is evaluated as:

$$z_R = \frac{\pi W_0^2}{\lambda} = \frac{\pi(19 \cdot 10^{-6}\ \text{m})^2}{1064 \cdot 10^{-9}\ \text{m}} = \frac{\pi \cdot 3.61 \cdot 10^{-10}}{1.064 \cdot 10^{-6}} \simeq 1.066 \cdot 10^{-3}\ \text{m} \sim 1\ \text{mm} \tag{7}$$

The beam waist expansion along the propagation direction is modeled by:

$$W(z) = W_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} \tag{8}$$

Evaluating the beam waist at the MOT position, located at a distance $d = 7$ mm from the focus ($z = 7$ mm):

$$W(7\ \text{mm}) = 19 \cdot 10^{-6}\ \text{m} \cdot \sqrt{1 + \left(\frac{7\ \text{mm}}{1\ \text{mm}}\right)^2} = 19 \cdot 10^{-6} \cdot \sqrt{50} \simeq 134.35\,\mu\text{m} \sim 126\,\mu\text{m} \tag{9}$$

This confirms that the physical parameters match the geometric propagation equations within laboratory alignment tolerances.

2.2 Trap Depth at the Fiber Tip and Kinematics

Since the total laser power P is conserved along z, the dipole potential scales inversely with the square of the beam waist ($U \propto 1/W^2$). The ratio of the trap depths satisfies:

$$\frac{U_{\text{tip}}}{U_{\text{MOT}}} = \frac{1/W_{\text{tip}}^2}{1/W_{\text{MOT}}^2} = \left(\frac{W_{\text{MOT}}}{W_{\text{tip}}}\right)^2 \tag{10}$$

Isolating U_{tip} and substituting our parameters:

$$U_{\text{tip}} = U_{\text{MOT}} \cdot \left(\frac{126\,\mu\text{m}}{19\,\mu\text{m}}\right)^2 \simeq U_{\text{MOT}} \cdot 43.98 \sim 6\,\mu\text{K} \cdot 44 \sim 250\,\mu\text{K} \tag{11}$$

Assuming the atoms undergo unhindered free fall over the vertical distance $d = 7$ mm, their classical kinetic energy equals the gravitational potential energy change ($E_k = U_{\text{grav}}$):

$$\frac{1}{2}mv^2 = mgd \implies v = \sqrt{2gd} = \sqrt{2 \cdot 9.81\ \text{m/s}^2 \cdot 7 \cdot 10^{-3}\ \text{m}} \simeq 0.37\ \text{m/s} = 37\ \text{cm/s} \tag{12}$$

3 Harmonic Approximation and Adiabaticity Analysis

3.1 Derivation of Radial Harmonic Frequency

The transverse intensity profile of the Gaussian beam is given by $I(r,z) = I_0(z)e^{-2r^2/W(z)^2}$. Thus, the localized radial dipole potential is:

$$U(r) = -U_0 e^{-\frac{2r^2}{W^2}} \tag{13}$$

Since the trapped atomic cloud is ultra-cold, the thermal energy restricts the atoms to the core of the beam, allowing us to assume $r \ll W$. Performing a Taylor expansion for $x \to 0$ ($e^{-x} \approx 1 - x$):

$$U(r) \sim -U_0 \left(1 - \frac{2r^2}{W^2}\right) = -U_0 + \frac{2U_0}{W^2} r^2 \tag{14}$$

Equating the quadratic term to the classical harmonic spring potential $V(r) = \frac{1}{2}kr^2$:

$$\frac{1}{2}kr^2 = \frac{2U_0}{W^2}r^2 \implies k = \frac{4U_0}{W^2} \tag{15}$$

The angular frequency of radial oscillations is therefore:

$$\omega_r = \sqrt{\frac{k}{m}} = \sqrt{\frac{4U_0}{mW^2}} \tag{16}$$

Using the value $U_0 = 6\,\mu\text{K} \cdot k_B \sim 8.28 \cdot 10^{-29}$ J and $m_{\text{Rb}} = 1.44 \cdot 10^{-25}$ kg:

$$\nu_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi \cdot 126 \cdot 10^{-6}}\sqrt{\frac{4 \cdot 8.28 \cdot 10^{-29}}{1.44 \cdot 10^{-25}}} = \frac{1}{7.916 \cdot 10^{-4}} \cdot \sqrt{2.3} \simeq 60.5 \text{ Hz} \tag{17}$$

The corresponding mechanical oscillation period is:

$$T_{\text{osc}} = \frac{1}{\nu_r} = \frac{1}{60.5 \text{ Hz}} \simeq 16.5 \text{ ms} \tag{18}$$

3.2 Rigorous Adiabaticity Parameter Verification

To ensure that the cloud does not undergo parametric excitation and heating during compression, the relative variation of the trap frequency within one cycle must satisfy $\frac{\Delta\omega}{\omega} \ll 1$. Formally:

$$\alpha = \left|\frac{1}{\omega_r^2}\frac{d\omega_r}{dt}\right| \ll 1 \tag{19}$$

Since $U_0 \propto 1/W^2$, the frequency scales as $\omega_r \propto \sqrt{U_0}/W \propto 1/W^2$. Given the waist profile:

$$\omega_r(z) = \frac{\omega_0}{1 + (z/z_R)^2} \tag{20}$$

Applying the chain rule for an atom falling under gravity ($z(t) = L - \frac{1}{2}gt^2$, where $L = 7$ mm and $v(t) = \frac{dz}{dt} = -gt$):

$$\frac{d\omega_r}{dt} = \frac{d\omega_r}{dz}\frac{dz}{dt} = \left(\omega_0 \cdot \frac{-2z/z_R^2}{(1 + (z/z_R)^2)^2}\right) \cdot (-gt) = \frac{2\omega_0 g z(t) t}{z_R^2(1 + (z/z_R)^2)^2} \tag{21}$$

Substituting this back into the expression for α:

$$\alpha(t) = \frac{(1 + (z/z_R)^2)^2}{\omega_0^2} \cdot \frac{2\omega_0 g z(t) t}{z_R^2(1 + (z/z_R)^2)^2} = \frac{2g z(t) t}{\omega_0 z_R^2} \tag{22}$$

Expanding $z(t)$ explicitly as a function of time:

$$\alpha(t) = \frac{2g}{\omega_0 z_R^2}\left(Lt - \frac{1}{2}gt^3\right) \tag{23}$$

At $t = 0$ (start of fall) and at the total transit time $t_{\text{fall}} = \sqrt{2L/g} \simeq 38$ ms (arrival at $z = 0$), we find $\alpha = 0$. To calculate the maximum risk of non-adiabatic loss, we find the stationary point by taking the time derivative:

$$\frac{d\alpha}{dt} = 0 \implies \frac{d}{dt}\left(Lt - \frac{1}{2}gt^3\right) = L - \frac{3}{2}gt^2 = 0 \implies t_{\max} = \sqrt{\frac{2L}{3g}} \tag{24}$$

Substituting $L = 7$ mm:

$$t_{\max} = \sqrt{\frac{2 \cdot 7 \cdot 10^{-3}}{3 \cdot 9.81}} \simeq 21.8 \text{ ms} \tag{25}$$

Evaluating $\omega_r(t_{\max}) \simeq 16732$ rad/s yields a maximum adiabatic parameter of:

$$\alpha_{\max} \sim 0.105 \tag{26}$$

Since $\alpha_{\max} \ll 1$, the compression behaves adiabatically throughout the entire transport.

3.3 Thermodynamic Filtering and Scaling

For an adiabatic process, the action integral $J = \frac{E}{\nu_r} = \text{const}$ is conserved. According to the equipartition theorem, the average energy scales with temperature ($E \propto k_B T$), which implies:

$$\frac{T}{\nu_r} = \text{const} \implies T \propto \nu_r \propto \frac{1}{W^2} \tag{27}$$

This indicates that compressing the cloud decreases spatial uncertainty Δx, which increases momentum uncertainty Δp and raises the temperature.

However, the initial MOT cloud at $T = 8\,\mu\text{K}$ cannot be captured entirely by a trap deep only $U_{\text{MOT}} = 6\,\mu\text{K}$. The ODT acts as a thermodynamic filter, trapping only the low-energy fraction of the Maxwell-Boltzmann distribution. The trapped sub-ensemble has a lower effective starting temperature of $T_{\text{captured}} \sim 1\,\mu\text{K}$. Under adiabatic compression, the final temperature at the tip scales as:

$$T_{\text{tip}} = T_{\text{captured}} \left(\frac{W_{\text{MOT}}}{W_{\text{tip}}} \right)^2 \sim 1\,\mu\text{K} \cdot 44 \sim 44\,\mu\text{K} \tag{28}$$

4 Phase-Space Capture Fraction in the Shallow Trap Limit

An atom at position $\vec{r}$ is successfully trapped if its transverse kinetic energy is less than the potential depth: $\frac{1}{2}m(v_x^2+v_y^2) < |U_{\mathrm{ODT}}(\vec{r})|$. The localized spatial density is $n_{\mathrm{MOT}}(\vec{r})$ and the velocity distribution is $f(\vec{v}) = (2\pi\mu^2)^{-3/2}e^{-v^2/2\mu^2}$, with $\mu^2 = \frac{k_BT}{m}$. The number of captured atoms is obtained by integrating over phase space, selecting only falling atoms ($v_z < 0$):

$$N_{\mathrm{ODT}} = \int d^3r\, n_{\mathrm{MOT}}(\vec{r}) \cdot \frac{1}{2}\int_0^{v_x^2+v_y^2<\frac{2|U|}{m}} dv_x dv_y\, \frac{1}{2\pi\mu^2}e^{-\frac{v_x^2+v_y^2}{2\mu^2}} \tag{29}$$

Transforming the velocity integrals to polar coordinates ($dv_xdv_y = \pi d(v_r^2)$) and applying the change of variables $x = \frac{v_r^2}{2\mu^2}$:

$$\int_0^{\frac{2|U|}{m}} \pi d(v_r^2)\frac{1}{2\pi\mu^2}e^{-\frac{v_r^2}{2\mu^2}} = \int_0^{\frac{|U|}{m\mu^2}} e^{-x}dx = \left[-e^{-x}\right]_0^{\frac{|U|}{k_BT}} = 1 - e^{-\frac{|U_{\mathrm{ODT}}(\vec{r})|}{k_BT}} \tag{30}$$

Therefore, the capture equation simplifies to:

$$N_{\mathrm{ODT}} = \int d^3r\, n_{\mathrm{MOT}}(\vec{r}) \cdot \frac{1}{2}\left(1 - e^{-\frac{|U_{\mathrm{ODT}}(\vec{r})|}{k_BT}}\right) \tag{31}$$

In the shallow trap limit ($|U_{\mathrm{ODT}}| \ll k_BT$), we can approximate $1 - e^{-x} \approx x$:

$$N_{\mathrm{ODT}} \simeq \int d^3r\, n_{\mathrm{MOT}}(\vec{r})\frac{|U_{\mathrm{ODT}}(\vec{r})|}{2k_BT} = \frac{U_0}{2k_BT}\int d^3r\, n_{\mathrm{MOT}}(\vec{r})\frac{W_0^2}{W^2(z)}e^{-\frac{2(x^2+y^2)}{W^2(z)}} \tag{32}$$

Given that the beam radius is much smaller than the cloud dimensions ($W(z) \ll R_x, R_y$), the transverse spatial integrals can be evaluated from $-\infty$ to $+\infty$:

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} e^{-\frac{2(x^2+y^2)}{W^2(z)}}dxdy = \frac{\pi W^2(z)}{2} \tag{33}$$

Substituting this back into the main expression causes $W^2(z)$ to cancel out:

$$N_{\mathrm{ODT}} = \frac{U_0}{2k_BT}\frac{\pi W_0^2}{2}\int_{-R_z}^{R_z} n_{\mathrm{MOT}}(0,0,z)dz \tag{34}$$

Modeling the MOT as a uniform ellipsoid of radii R_x, R_y, R_z with constant density $n_{\mathrm{MOT}} = \frac{3N_{\mathrm{MOT}}}{4\pi R_xR_yR_z}$:

$$N_{\mathrm{ODT}} = \frac{U_0\pi W_0^2}{4k_BT}\left(\frac{3N_{\mathrm{MOT}}}{4\pi R_xR_yR_z}\cdot 2R_z\right) = \frac{3}{8}\frac{U_0W_0^2}{k_BTR_xR_y}N_{\mathrm{MOT}} \tag{35}$$

This confirms that the capture efficiency $\frac{N_{\mathrm{ODT}}}{N_{\mathrm{MOT}}} = \frac{3}{8}\frac{U_0W_0^2}{k_BTR_xR_y}$ is independent of the loading distance d, and independent of the waist size W_0 because the product $U_0W_0^2 \propto P$ depends only on the laser power.

5 Conveyor Belt Moving Lattice Dynamics

5.1 Interference Pattern and Potential Depth

We now implement an active transport mechanism using two counter-propagating lasers with powers $P_1 = 1$ W and $P_2 = 1.5$ W. The interference of the electric fields scales as $I_{\text{peak}} \propto (\sqrt{P_1} + \sqrt{P_2})^2 = (1 + \sqrt{1.5})^2 \simeq 4.95$.

Accounting for non-ideal beam overlap in the laboratory, we assume a conservative amplification factor of ~ 4. The scaled trap depths are:

- $U_{\text{MOT}} \sim 6\,\mu\text{K} \cdot 4 = 25\,\mu\text{K}$
- $U_{\text{tip}} \sim 25\,\mu\text{K} \cdot 44 = 1.2$ mK

5.2 Derivation of the Transport Velocity via Prostaferesi

The two fields traveling in opposite directions with a slight frequency offset ($\omega_1 - \omega_2 = 2\pi\Delta\nu$) are expressed as:

$$E_1 = E_0 \cos(kz - \omega_1 t), \quad E_2 = E_0 \cos(-kz - \omega_2 t) = E_0 \cos(kz + \omega_2 t) \tag{36}$$

Applying the trigonometric identity $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$:

$$E_{\text{tot}} = 2E_0 \cos\left(\frac{2kz - (\omega_1 - \omega_2)t}{2}\right) \cos\left(\frac{(\omega_1 + \omega_2)t}{2}\right) \tag{37}$$

$$E_{\text{tot}} = 2E_0 \cos\left(kz - \frac{\omega_1 - \omega_2}{2} t\right) \cos\left(\frac{\omega_1 + \omega_2}{2} t\right) \tag{38}$$

The atoms average out the rapid optical oscillation term $\cos\left(\frac{\omega_1+\omega_2}{2} t\right)$. The effective potential is proportional to the mean square intensity of the slow spatial envelope:

$$U(z,t) = -U_0 \cos^2\left(kz - \frac{2\pi\Delta\nu}{2} t\right) = -U_0 \cos^2\left(\frac{2\pi}{\lambda}\left(z - \frac{\lambda\Delta\nu}{2} t\right)\right) \tag{39}$$

This defines a periodic lattice structure moving at a constant velocity of:

$$v = \frac{\lambda\Delta\nu}{2} \tag{40}$$

To match the average free-fall velocity $v = 0.184$ m/s required to cross 7 mm in 38 ms:

$$\Delta\nu = \frac{2v}{\lambda} = \frac{2 \cdot 0.184\ \text{m/s}}{1064 \cdot 10^{-9}\ \text{m}} \simeq 345.86\ \text{kHz} \sim 346\ \text{kHz} \tag{41}$$

The full three-dimensional trapping potential combining the transverse Gaussian confinement and longitudinal interference nodes is given by:

$$U(r,z) = -U_0 e^{-\frac{2r^2}{W^2}} \cos^2\left[\frac{2\pi}{\lambda}\left(z - \frac{\lambda\Delta\nu}{2} t\right)\right] \tag{42}$$

6 Axial vs. Radial Frequency Scaling and Cloud Geometry

6.1 Longitudinal Harmonic Oscillator Approximation

We model the potential well around an interference minimum ($z \to 0$). Performing a Taylor expansion on the cosine function ($\cos(x) \approx 1 - \frac{x^2}{2}$):

$$\cos^2(kz) \approx \left(1 - \frac{(kz)^2}{2}\right)^2 \approx 1 - k^2 z^2 \tag{43}$$

Substituting this expansion into the longitudinal potential expression:

$$U(z) = -U_{\text{ax}}(1 - k^2 z^2) = -U_{\text{ax}} + U_{\text{ax}} k^2 z^2 \tag{44}$$

Equating this to a 1D harmonic spring potential $V(z) = \frac{1}{2} k_z z^2$:

$$\frac{1}{2} k_z z^2 = U_{\text{ax}} k^2 z^2 \implies k_z = 2 U_{\text{ax}} k^2 \tag{45}$$

The axial frequency ω_z is given by:

$$\omega_z = \sqrt{\frac{2 U_{\text{ax}} k^2}{m}} = k \sqrt{\frac{2 U_{\text{ax}}}{m}} \implies \nu_z = \frac{1}{\lambda} \sqrt{\frac{2 U_{\text{ax}}}{m}} \tag{46}$$

Crucially, the axial frequency depends only on the potential depth and wavelength, with no dependency on the beam waist W.

Evaluating ν_z at the MOT position ($U_{\text{ax}} = 25\,\mu\text{K} \cdot k_B \sim 3.45 \cdot 10^{-27}$ J):

$$\nu_{z,\text{MOT}} = \frac{1}{1064 \cdot 10^{-9}} \sqrt{\frac{2 \cdot 3.45 \cdot 10^{-27}}{1.44 \cdot 10^{-25}}} \simeq 70.5 \text{ kHz} \tag{47}$$

Since U_{ax} increases by a factor of 44 at the fiber tip, the axial frequency scales as $\nu_z \propto \sqrt{U_{\text{ax}}}$, meaning it increases by a factor of $\sqrt{44} \simeq 6.63$:

$$\nu_{z,\text{tip}} = 70.5 \text{ kHz} \cdot 6.6 \simeq 450 \text{ kHz} \tag{48}$$

Trap Parameter	**MOT Position**	**Fiber Tip**	**Scaling Factor**
Radial Depth (U_{rad})	$\sim 25\,\mu$K	~ 1.2 mK	44
Axial Depth (U_{ax})	$\sim 25\,\mu$K	~ 1.2 mK	44
Radial Frequency (ν_r)	~ 120 Hz	~ 5 kHz	44
Axial Frequency (ν_z)	~ 70 kHz	~ 450 kHz	6.6

Table 1: Comprehensive parameter evolution during conveyor belt transport.

6.2 Cloud Spatial Evolution and Quantum Occupancy

The thermal width of a classical cloud is governed by $\sigma = \sqrt{\frac{k_B T}{m\omega^2}}$. Under adiabatic transport ($T \propto \nu$), the width scales as $\sigma \propto \frac{1}{\sqrt{\nu}}$. Evaluating the cloud dimensions at the MOT position and at the fiber tip yields:

- **MOT Position:** $\Delta\sigma_{\text{rad}} \sim 130\,\mu$m, $\Delta\sigma_{\text{ax}} \sim 220$ nm
- **Fiber Tip:** $\Delta\sigma_{\text{rad}} \sim 18\,\mu$m, $\Delta\sigma_{\text{ax}} \sim 95$ nm (highly flattened pancake geometry)

The average quantum state occupancy $\langle m \rangle$ is calculated using the Bose-Einstein distribution function:

$$\langle m \rangle = \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \tag{49}$$

At the fiber tip, with $T_r \sim 350\,\mu\text{K}$ and $\nu_r \sim 5$ kHz, the radial axis yields:

$$\frac{h\nu_r}{k_B T_r} \sim 0.0008 \implies \langle m_r \rangle = \frac{1}{e^{0.0008} - 1} \simeq 1250 \quad \text{(Highly classical)} \tag{50}$$

Along the axial direction, with $T_z \sim 52.8\,\mu\text{K}$ and $\nu_z \sim 450$ kHz:

$$\frac{h\nu_z}{k_B T_z} \sim 0.4 \implies \langle m_z \rangle = \frac{1}{e^{0.4} - 1} \simeq 1.89 \quad \text{(Approaching the ground state)} \tag{51}$$

7 EIT Sideband Cooling and Lamb-Dicke Parameter

7.1 Calculation of the Lamb-Dicke Parameter (η)

To implement Electromagnetically Induced Transparency (EIT) sideband cooling at $\lambda_{\mathrm{EIT}} = 780$ nm, the system must be evaluated relative to the Lamb-Dicke limit. The absorption wavenumber is $k = \frac{2\pi}{780 \cdot 10^{-9}\ \mathrm{m}} \simeq 8.055 \cdot 10^6\ \mathrm{m}^{-1}$. The harmonic ground-state wavepacket size $x_0 = \sqrt{\frac{\hbar}{2m\omega_z}}$ along the axial direction ($\omega_z = 2\pi \cdot 450$ kHz $\simeq 2.9 \cdot 10^6$ rad/s) is:

$$x_{0,\mathrm{ax}} = \sqrt{\frac{1.054 \cdot 10^{-34}\ \mathrm{J \cdot s}}{2 \cdot 1.44 \cdot 10^{-25}\ \mathrm{kg} \cdot 2.9 \cdot 10^6\ \mathrm{rad/s}}} \simeq 1.10 \cdot 10^{-8}\ \mathrm{m} = 11\ \mathrm{nm} \tag{52}$$

The axial Lamb-Dicke parameter is calculated as:

$$\eta_{\mathrm{ax}} = k \cdot x_{0,\mathrm{ax}} = 8.055 \cdot 10^6\ \mathrm{m}^{-1} \cdot 1.10 \cdot 10^{-8}\ \mathrm{m} \simeq 0.088 \sim 0.09 \tag{53}$$

Since $\eta_{\mathrm{ax}} \ll 1$, the atomic transport axis is firmly within the Lamb-Dicke regime, ensuring that photon recoil shifts are heavily suppressed during EIT cooling cycles.

7.2 Differential AC Stark Shifts inside the ODT

The intense 1064 nm trap shifts the atomic energy levels. Using the frequency conversion factor $1\,\mu\mathrm{K} \cdot k_B \sim 20.836\ \mathrm{kHz} \cdot h$, we calculate the energy shifts at the center of the 1.2 mK trap:

- **Ground State Shift ($\Delta\nu_g$):** The ground state experiences an attractive potential:

$$\Delta\nu_g = -1200\,\mu\mathrm{K} \cdot 20.836\ \mathrm{kHz}/\,\mu\mathrm{K} \simeq -25.0\ \mathrm{MHz} \tag{54}$$

- **Excited State Shift ($\Delta\nu_e$):** At 1064 nm, the excited state $5P_{3/2}$ has a repulsive polarizability relative to the ground state, with a scalar ratio of ~ -1.67:

$$\Delta\nu_e = -1.67 \cdot \Delta\nu_g = -1.67 \cdot (-25\ \mathrm{MHz}) \simeq +41.8\ \mathrm{MHz} \sim +45\ \mathrm{MHz} \tag{55}$$

The net differential AC Stark shift affecting the single-photon resonance frequency is:

$$\Delta\nu_{\mathrm{eff}} = \Delta\nu_e - \Delta\nu_g \simeq 45\ \mathrm{MHz} - (-25\ \mathrm{MHz}) = 70\ \mathrm{MHz} \tag{56}$$

To restore single-photon resonance conditions inside the trap core, the cooling and probe lasers must be detuned by approximately +70 MHz. Because both hyper-fine ground states ($F = 1$ and $F = 2$) shift by the exact same amount ($\Delta\nu_g$), the two-photon resonance condition remains unaffected, protecting the coherence of the EIT dark state.

7.3 Inhomogeneous Broadening Profile

Because the radial temperature is $T_r \sim 350\,\mu$K, atoms sample outer regions of the 1.2 mK potential well, encountering variations in intensity. The explored fraction of the trap depth is $\frac{T_r}{U_0}$. The single-photon disomogeneous broadening is given by:

$$\delta\nu_{\mathrm{inh}} \sim \Delta\nu_{\mathrm{max}} \cdot \frac{T_r}{U_0} = 70\ \mathrm{MHz} \cdot \left(\frac{350\,\mu\mathrm{K}}{1200\,\mu\mathrm{K}}\right) \simeq 20.4\ \mathrm{MHz} \sim 20\ \mathrm{MHz} \tag{57}$$

While this 20 MHz broadening broadens the single-photon background transition, the two-photon EIT Fano resonance is a common-mode configuration and is isolated from this spatial broadening. Its linewidth is restricted to $\sim$ 1 MHz, as observed in Choi et al., dominated primarily by magnetic state blending and intrinsic laser phase noise.